# Project Milestones

| Phase 1 — Foundation & Database | |
|---|---|
| 1 | Full folder structure created |
| 2 | .env and config.py configured |
| 3 | Supabase migration SQL written (all 4 tables) |
| 4 | supabase_client.py with basic read/write helpers |
| 5 | DB connection verified with test script |

| Phase 2 — Input Parsing | |
|---|---|
| 1 | Sample books_input.xlsx created with all required columns |
| 2 | parse_excel.py service written |
| 3 | FastAPI route POST /parse-input implemented |
| 4 | Validated row inserted into books table in Supabase |
| 5 | Book ID returned in response |

| Phase 3 — Prompt Builder & OpenAI Service | |
|---|---|
| 1 | prompt_builder.py with outline, chapter, and summary prompts |
| 2 | openai_service.py wrapper with retry logic |
| 3 | Both modules tested independently with dummy data |

| Phase 4 — Outline Generation (Stage 1 Logic) | |
|---|---|
| 1 | outline_service.py with gating checks |
| 2 | FastAPI route POST /outline/generate |
| 3 | FastAPI route POST /outline/regenerate |
| 4 | Outline stored with version number in Supabase |
| 5 | Book status updated correctly at each gate |

| Phase 5 — Chapter Generation (Stage 2 Logic) | |
|---|---|
| 1 | chapter_service.py with summary-chaining logic |
| 2 | FastAPI route POST /chapters/generate |
| 3 | FastAPI route POST /chapters/regenerate/{chapter_id} |
| 4 | Each chapter and auto-summary stored in Supabase |

| | |
|---|---|
| 5 | Per-chapter gate logic applied (yes / no / no_notes_needed) |

| Phase 6 — Notification Service | |
|---|---|
| 1 | notification_service.py written |
| 2 | SMTP email setup with templates |
| 3 | MS Teams webhook integrated |
| 4 | Notifications hooked into outline and chapter services |

| Phase 7 — Final Compilation (Stage 3 Logic) | |
|---|---|
| 1 | docx_builder.py using python-docx |
| 2 | pdf_builder.py converting docx to PDF |
| 3 | compilation_service.py with final gate checks |
| 4 | FastAPI route POST /compile |
| 5 | Output files uploaded to Supabase Storage |
| 6 | Editor notified with download link |

| Phase 8 — n8n Workflows | |
|---|---|
| 1 | outline_workflow.json — Excel trigger, FastAPI calls, status gating |
| 2 | chapter_workflow.json — chapter loop, review flag handling |
| 3 | compilation_workflow.json — final gate, compile trigger, notification |
| 4 | All three workflows tested end to end |

| Phase 9 — Docker & Local Dev Setup | |
|---|---|
| 1 | docker-compose.yml for n8n and FastAPI |
| 2 | Dockerfile for FastAPI service |
| 3 | .env.example with all required keys |
| 4 | README.md with setup, demo, and architecture explanation |

| Phase 10 — Polish & Submission Prep | |
|---|---|
| 1 | All code cleaned up, dead code removed |
| 2 | Sample output .docx and PDF generated |
| 3 | DB structure screenshots taken |

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